

12.3 Latent Diffusion and Text Conditioning

12× fewer latent cells. CFG: $s=1$ is still conditional.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Pixel-space diffusion on 256×256 RGB is expensive. **Latent diffusion** (Rombach et al.) denoises in a compressed space. **Text** enters as a condition, usually from a frozen CLIP text tower (Week 5).

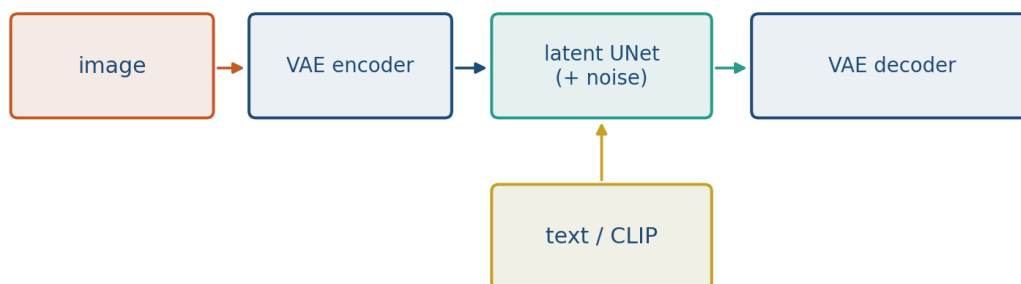
First time this appears. **Latent diffusion** denoises in a **VAE latent**, not in pixels. **CFG** mixes a conditional and an unconditional noise prediction.

What. Encode $x_0 \rightarrow z_0$ with a frozen VAE encoder, diffuse z , decode at the end. Text: condition on a CLIP (or T5) vector. CFG scale s sharpens the condition. **Why.** Pixels are huge. Latent cells are fewer (on the order of 12× fewer in the Stable Diffusion cartoon). Text needs a condition or you sample generic images. **Architecture.** VAE enc/dec (frozen) + denoiser on z_t + text encoder. CFG: two forwards, $\hat{\epsilon} = \epsilon_u + s(\epsilon_c - \epsilon_u)$. **How.** $s = 1$ is still conditional (not "off"). $s = 0$ is the uncond branch. High s : sharper, less diverse. **Formula.** $\hat{\epsilon}_\theta = \epsilon_\theta(z_t, t, \emptyset) + s(\epsilon_\theta(z_t, t, c) - \epsilon_\theta(z_t, t, \emptyset))$. **Tradeoffs.** + Cheaper than pixel DDPM, steerable with text. – VAE artifacts; CFG is a quality–diversity knob; still many reverse steps.

1. Diffuse in latent space

A pretrained encoder E maps an image to $z_0 = E(x)$. You run the forward process of note **12.1** on z , not on pixels. A decoder D_{VAE} turns the final latent back into an image. Training the U-Net on latents is cheaper; the VAE already threw away some high-frequency detail.

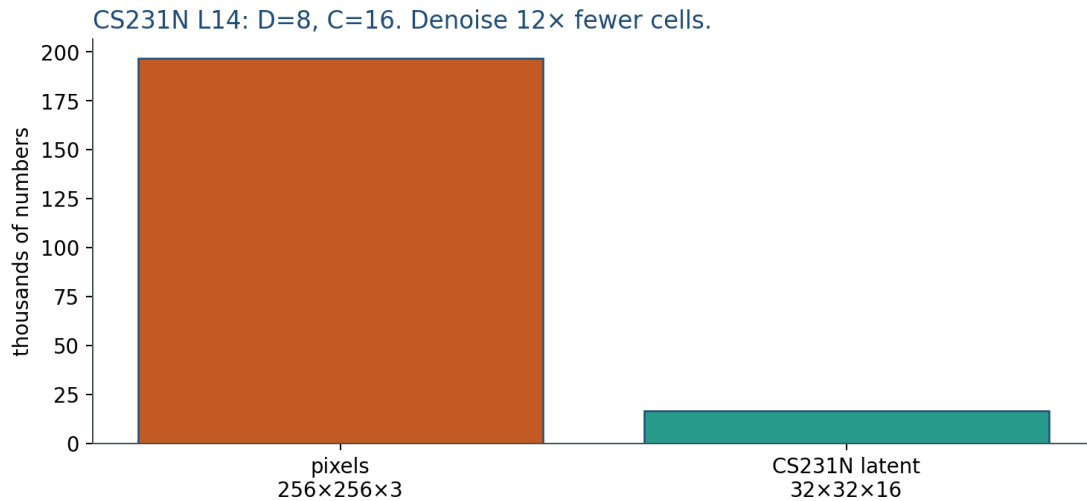
Latent diffusion: denoise a cheap latent, condition on text



Stable Diffusion is this picture plus a big text-conditioned U-Net. The idea is the same at homework scale: compress, denoise, decode.

Typical sizes to say out loud: an image $512 \times 512 \times 3$ becomes a latent like $64 \times 64 \times 4$. You pay diffusion steps on the small tensor.

A classroom setting is $D = 8$, $C = 16$: $256 \times 256 \times 3 = 196,608$ numbers versus $32 \times 32 \times 16 = 16,384$ (about $12 \times$ fewer). Stable Diffusion's public $64 \times 64 \times 4$ on 512^2 is about $48 \times$. Same slogan: denoise the small tensor.



The same lecture notes that modern LDM pipelines stack a VAE, a GAN (for the decoder), and diffusion. DiT exists. This hour is the VAE-latent picture, not a transformer-block homework.

2. Conditioning with CLIP text

You want $p(x \mid \text{caption})$, not an unconditional generator. Encode the caption with CLIP's **text** tower (note 5.1). Inject that vector by:

- concatenating it to the U-Net's timestep embedding, or
- **cross-attention**: image (or latent) tokens query the text tokens.

Classifier-free guidance trains with and without the text, then at sample time does

$$\hat{\epsilon} = \epsilon_{\theta}(z_t, t, \emptyset) + s \left(\epsilon_{\theta}(z_t, t, c) - \epsilon_{\theta}(z_t, t, \emptyset) \right).$$

Scale $s > 1$ sharpens adherence to the prompt and can wash out diversity. Report s .

CLIP scores pairs; it does not decode a caption (note 5.2). Here you only need the text embedding as a handle.

3. What to measure

Text-to-image is not "the picture looks nice." Report a prompt-alignment score (CLIP cosine of image vs text), a coverage or diversity check, and a failure case (wrong count, wrong object, reading text in the image). Conditioning does not remove Week 5's bias story: the same CLIP geometry is in the loop.

4. Teaching this note

About **35 minutes** at the board, then **~12 minutes** of selected video from the first 45 minutes (do not play the five-hour coding rest).

- **0–12 min.** Why latents: VAE encode / denoise / decode. Count pixels vs latent cells.
 - **12–22 min.** CLIP text tower as c . Cross-attention in one sentence.
 - **22–33 min.** Classifier-free guidance with numbers. Diversity vs prompt match.
 - **Then** play Umar Jamil Stable Diffusion **4:30–12:07** (what SD is; forward/reverse in this stack) and **22:20–31:00** (CFG). Optional 31:00–39:54 (CLIP + VAE + text-to-image) if time. Stop by **45:00**.
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5. Worked example

Latent size. A $512 \times 512 \times 3$ image is 786,432 numbers. A $64 \times 64 \times 4$ latent is 16,384 numbers: about $48 \times$ fewer. Fifty reverse steps on the latent are cheaper than fifty steps on pixels.

Guidance. Let the net's two noise guesses (one scalar, to see the algebra) be $\varepsilon_{\theta}(z_t, t, \emptyset) = 0.8$ (unconditional) and $\varepsilon_{\theta}(z_t, t, c) = 0.2$ (with prompt c). The difference is $0.2 - 0.8 = -0.6$.

$$\hat{\varepsilon} = 0.8 + s(-0.6) = 0.8 - 0.6s.$$

s	$\hat{\varepsilon}$	reading
1	0.2	just the conditional net
3	-1.0	step farther along the prompt direction
7.5	-3.7	very sharp prompt match; variety usually drops

$s = 1$ is “use the text-conditioned prediction.” $s > 1$ **extrapolates** away from the unconditioned guess. That is why a huge s can look like a poster of the prompt and still miss modes (Week 11's coverage lesson, now in prompt space).

 CFG scale $\varepsilon_{\theta}(z_t, t, c)$=

$s=1,3,7.5$ on a scalar pair" src="graphics/12.3-latent-conditioning/cfg-scale.png" />

Randomly **drop** the text in training so one net is both conditional and unconditional. Two forwards at sample time, then mix with s .

6. Where students get stuck

- Fine-tuning CLIP's image tower at sample time. Text-to-image needs the **text** tower (and a VAE decoder). The image tower is for scoring pairs, not for sampling.
- Calling $s = 1$ “no text.” $s = 1$ is still conditional; the empty prompt $\emptyset$ is the other call.

- Measuring only FID or only “looks nice,” with no prompt-alignment number and no failure case.
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7. Video

Watch [Umar Jamil, Coding Stable Diffusion from scratch in PyTorch](#), **0:00–45:00** assigned, with in-class pauses at **4:30–12:07** (what Stable Diffusion is; forward/reverse) and **22:20–39:54** (classifier-free guidance, CLIP, VAE, text-to-image).

Do not start the coding chapters (VAE implementation from 44:30 onward) in this lecture. The board is the picture; the lab is still 2-D denoising from notes **12.1–12.2**.

8. Practice

1. Why denoise latents instead of pixels if you already have a VAE?
2. CLIP’s image tower is optional at sample time for text-to-image. Which CLIP tower is not?
3. Guidance scale $s = 1$ versus $s = 12$: what do you expect to happen to prompt match and to variety?
4. $\epsilon_{\text{uncond}} = 1.0$, $\epsilon_{\text{cond}} = 0.4$, $s = 5$. Compute $\hat{\epsilon}$.
5. A $256 \times 256 \times 3$ image vs an $32 \times 32 \times 4$ latent: how many numbers in each, and what is the ratio (image / latent)?